

HIKET — the storyline, v3

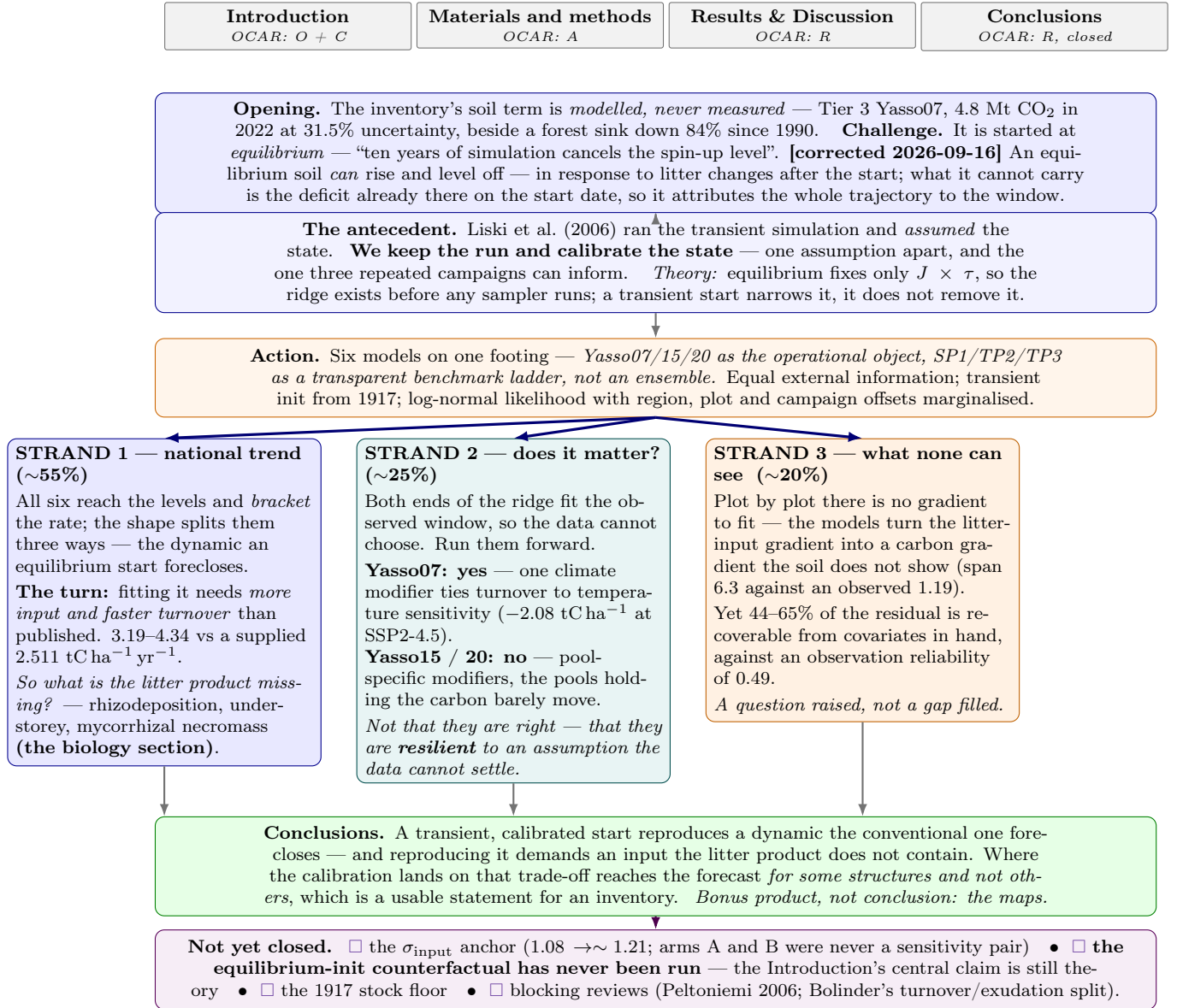
second pass: the arc fitted to the structure of the paper

Version of October 7, 2026. Numbers keyed to RUN_IDs 20260831_1624* (arm B). Built from the 19 annotations on v2 (revisions/HIKET_storyline_v2_annotated_20260903.pdf).

How to read this. The outer shell is now the conventional one — Introduction, Materials and methods, Results and Discussion, Conclusions — because that is what the paper has to be. **OCAR has not been abandoned; it has been moved inside.** The Opening and Challenge live in the Introduction, the Action is the Methods, and the Resolution is distributed across the Results in the order of importance set below. Each block is still one claim, its evidence, and nothing else. ▶ **name** marks a figure; ★ **TO BUILD: name** one still to build; □ **LIT REVIEW: topic** a literature review that must happen before the section can be written. **New in this revision:** a grey ⊢ REFS line under a claim lists the sources it rests on — ✓ verified against the source, □ still to check. The evidence tables and draft text behind them live in `literature/NN_topic/README.md`. Amber boxes are things that could sink a claim; green boxes are tests we could run; **violet boxes carry Lorenzo’s v2 annotations** and what was done about them.

national trend — the main story ~55%	ridge position ~25%	local variance ~20%
once the start is estimated from the trajectory, where does it place the models on the input/turnover exchange, and how far does the trajectory narrow it?	does the position on the ridge reach the extrapolation, and for which structures?	what the models cannot see at all — raised as questions, not solved

Annotation (p17): “*your draft is a lot on the model perspective but it will have to be more anchored in biology in the end ... weight is about 50–60% general trends, 20–30% this side arc ... remaining unexplained local variance and what we could do about it (very briefly the last).*” **Done:** the budget above governs this version, and the biological content that the input result demands is now §3.0.9 rather than four lines in a mechanism list.



Contents

1	Introduction	3
1.1	A sink that appears to be levelling off	3
1.2	Why it matters for the national greenhouse gas balance	4
1.3	The convention: soils initialised at steady state	5
1.4	Finnish forests are a managed landscape, not an equilibrium one	6
1.5	Objectives	8
2	Materials and methods	8
2.1	Data	8
2.2	The model ladder	8
2.3	Transient initialisation	10

2.4	A fair frame	10
2.5	Calibration and error model	11
2.6	Analyses	11
3	Results and discussion	11
	Strand 1 — the national trajectory (55%)	11
	Strand 2 — does the starting-state assumption reach the warming response? (25%)	22
	Strand 3 — what none of the models can see (20%)	24
4	Conclusions	26
	Open decisions and what is still missing	28

1 Introduction

1.1 A sink that appears to be levelling off

1.1.0.1 What the inventory measures. On the balanced plot set the mean stock runs 64.5 → 71.5 → 73.6 tC ha⁻¹ across the three campaigns: +7.0, then +2.1. Accumulation, bending over. ► [F3_mean_SOCalt](#)

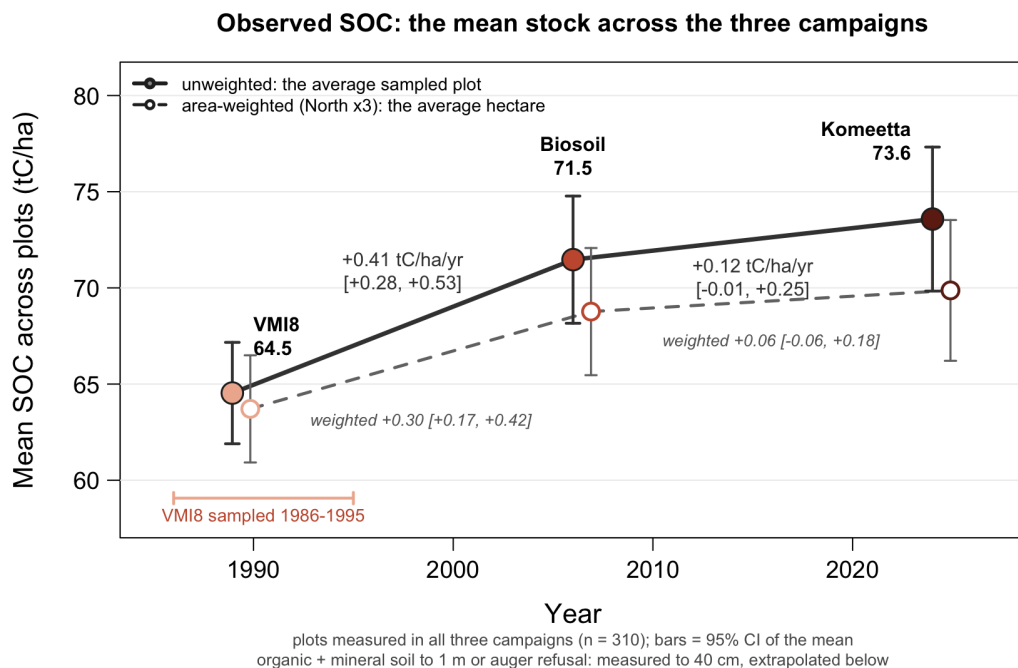


Figure 1: F3_mean_SOCalt

Opening figure, deliberately model-free (2026-09-04). v3 opened on F3_mean_soc, which draws six model trajectories through the campaign means — a Results answer shown before the question. F3_mean_SOCalt is the same data on the same basis with the models removed: three campaign means, the two segments between them, and their slopes. The model version now appears where it belongs, at the first Results claim.

1.1.0.2 The rate, and what it is a rate of. $+0.259 \pm 0.040 \text{ tC ha}^{-1} \text{ yr}^{-1}$ over 1985–2024 on the balanced set, whole profile, true observation years. Over the last interval alone it is $+0.117 \pm 0.066$ — a confidence interval that includes zero.

Which of these two numbers carries which claim — state it, and never let them swap. The **full span** (+0.259) is what the models are judged against, and it is the robust one: it survives a shared campaign-level offset of up to $\sim 9\%$. The **last interval** (+0.117) is fragile — it dissolves at $\tau_C \approx 2\%$, and no field replication exists to measure that offset. ★ **The uncomfortable part:** the levelling-off, which is the paper’s opening hook, is the *contrast* between the two intervals — so the hook rests partly on the fragile number even though the model comparison rests on the robust one. Say so where the hook is introduced, not only in the limitations. Memory **campaign-comparability-limits-the-trend**.

Annotation (p3): “Which rate? And which report?” Fixed here: the basis is now stated wherever a rate appears (balanced set $n = 310$, unweighted, whole profile, true years), because the same data give +0.209 on the pairwise set — a factor 1.8 on the headline. The reporting basis was decided on 2026-08-17 and every quoted rate in the paper must name it.

1.1.0.3 Why the levelling-off is the interesting part. A soil that is still accumulating and a soil that has stopped are the same stock and different futures. The second derivative, not the stock, is what an inventory projection turns on.

Do not present the observed rate as truth (annotation, p7). Three campaigns measured by three protocols over four decades give a rate with a real and partly unquantified comparability error: 1985 is a decade wide, its organic layer had to be reconstructed, and the short interval’s CI includes zero. The paper states what was measured and how, and treats agreement between models and observations as consistency rather than as validation.

1.2 Why it matters for the national greenhouse gas balance

1.2.0.1 The soil term is produced by a model, not measured. Finland reports forest mineral-soil carbon with a Tier 3 method: inventory data and biomass models supply a litter input, Yasso07 converts it to an annual stock change for dead organic matter and soil organic matter combined, to one metre, in two regions. No soil flux is measured anywhere in the chain.

└ REFS ✓ Statistics Finland NID 2024 §6.4 (method in force; “*builds on the research by Liski et al. (2006)*”); ✓ Liski et al. 2006 (the method’s founding paper)

1.2.0.2 And it is large enough to decide the sign. In 2022 mineral forest soils were reported as a sink of 4.8 Mt CO₂ beside a living-biomass sink of 12.6, while the forest land sink had fallen 84% since 1990 and the LULUCF sector had become a net source of 4.4 Mt CO₂ eq. The soil term carries 31.5% uncertainty against 16.3% for tree biomass, and Forest Land’s combined figure is 83.9%.

└ REFS ✓ NID 2024 §6.1, §6.4.3 (all figures); ✓ Lehtonen & Heikkinen 2015 (source of the 31.5% — and of our σ_{input} prior centre, which is worth saying once)

1.2.0.3 ★ The chain states its own initialisation assumption, and that is the proposition this paper tests. The inventory fixes the initial soil state by a spin-up to steady state — on NFI6 (1971–76) litter and 1960–90 mean weather — justified in one sentence: “*approximately ten years of simulation since spin-up is enough to cancel out the effect of the spin-up level*”. We test that over a 35-year observed window with the initial state calibrated rather than assumed.

└ REFS ✓ NID 2024 §6.4 (verbatim); □ Peltoniemi et al. 2006, For. Ecol. Manage. 232:75–85 — the paper it cites, **read before quoting**

Tone. Quote the *inventory* citing Peltoniemi et al., not Peltoniemi et al. directly: the 2006 claim is probably narrower than the use made of it, and Mikko is a close collaborator. The sentence is a foil for a *convention*, never for a colleague.

1.2.0.4 What the inventory asks a model to do. Not to rank plots, but to reproduce a national trajectory over decades and project it forward. That is the quantity this paper evaluates, and saying so early prevents a later result (near-zero plot-level skill) from reading as failure.

1.3 The convention: soils initialised at steady state

1.3.0.1 It is the standard across the lineages used for forest soil reporting. Yasso in the Finnish inventory (10 000-year run to steady state), RothC — where the equilibrium run additionally back-solves the annual carbon input from the measured stock — and the spin-ups of global land models. The instructive exception is Canada’s CBM-CFS3, which initialises through repeated rotations of growth and disturbance and so reports a deliberately non-equilibrium state.

└ REFS ✓ Lehtonen et al. 2016 (“*running ... into a steady state*”, and its premise: “*if we assume that this average level of inputs and climate has remained steady over centuries*”); ✓ NID 2024 §6.4; ✓ Contreras et al. 2026 (RothC practice); □ Kurz et al. 2009 (CBM-CFS3, secondary so far); □ Parton et al. 1987 (Century)

1.3.0.2 ★ And Liski et al. (2006) is the antecedent: they ran the transient simulation and assumed the state. Same country, same model family, same inventory-derived litter, the same idea of following the management history forward — “*the soil and litter carbon pools at the beginning of the study period were calculated by assuming a steady state ... starting from this steady state in 1922, the model was run using annually varying values of litter input and temperature.*” This paper keeps the run and **calibrates the state**. One assumption apart, and it is the one repeated campaigns can inform.

└ REFS ✓ Liski et al. 2006, Ann. For. Sci. 63:687–697 (verbatim, §2)

Lineage, not correction (Lorenzo, 2026-09-04: “*that’s the most important reference*”). Aleksi and Mikko are coauthors on it. The paper evolves the group’s own method; frame it collegially throughout, and give Liski the position in the Introduction that this draft previously gave to a passing σ_{input} comparison.

1.3.0.3 The convention has been criticised before — what has been missing is the test. The equilibrium assumption is known to bias the calibration, not only the initial state: fitting to current stocks under it overestimates the decay rate of the slowest pool, and spin-up overestimates recently disturbed sites. Where it has been relaxed, the replacement state has been *imposed* — from an assumed disturbance history, a thermal proxy, or a prescribed correction — rather than inferred from observed change in the stock.

└ REFS ✓ Wutzler & Reichstein 2007, Biogeosciences 4:125–136 (with Yasso; small accumulation makes implied equilibrium stocks “*virtually approach infinity*”); ✓ Carvalhais et al. 2008 (relaxing it: model efficiency +21%, biased parameters removed); ✓ Kanari et al. 2022; ✓ Lee & Viscarra Rossel 2020 (−2.0 to −4.5 Mg C ha^{−1} from initialisation alone); ✓ Le Noë et al. 2023 (~250 models, little independent time-series validation)

“Near-universal and rarely examined” has been cut, and must not come back. It is checkable and wrong: Wutzler & Reichstein made essentially this argument in 2007, using Yasso. The defensible novelty is narrower and still a first — **the initial state calibrated against repeated national soil inventories**, which is what the campaigns now allow.

1.3.0.4 What steady state assumes. In all of these models a pool loses carbon in proportion to how much it holds, and nothing else. Under a constant input and a constant climate such a system drifts to one stock and stays there: input multiplied by mean residence time, the familiar $C^* = J/k$. Starting a simulation at that stock is not an assumption about the soil. It is an assumption about its *history* — that input and climate have been effectively unchanged for long enough that even the slowest pool has caught up, and that nothing has disturbed it in between.

Annotation (p6): “*‘donor-controlled’ sounds jargon in this context, be simpler.*” Done — the property is now stated in words (“a pool loses carbon in proportion to how much it holds”). The term itself belongs in the Methods, once, if at all.

⊢ REFS ✓ Olson 1963, Ecology 44:322–331 ($X = L/k$); □ Ågren & Bosatta 1998 or Manzoni & Porporato 2009 (general form — pick one); □ Sierra et al. 2017 (age/turnover/transit vocabulary, to keep our terms exact)

1.3.0.5 Two properties follow, and the paper turns on both. The equilibrium condition fixes only the *product* $J \times \tau$, so a level alone can never separate a fast soil fed much litter from a slow soil fed little — **the ridge is derivable from the model definition, before any sampler runs**. And the stationary state is a function of the parameters alone, so an equilibrium start has no distance left to travel: it can track a moving target, but it cannot express a soil still catching up.

⊢ REFS ✓ derivation, ours (literature/03_equilibrium_theory); □ Luo et al. 2017, Biogeosciences 14:145–161 (storage *capacity* vs *potential* — adopt the vocabulary only after checking their definitions)

1.3.0.6 And what follows from it. An equilibrium start is already at the end of the accumulation. It cannot produce a rise, so it cannot show a rise levelling off; the very dynamic the inventory needs is the one the convention defines away.

1.4 Finnish forests are a managed landscape, not an equilibrium one

1.4.0.1 The history is the argument. Slash-and-burn across most of the productive forest land of the east, woodland grazing suppressing regeneration, tar burning and high-grading: by the first inventory, old trees were effectively absent from southern Finland outside the least accessible areas and forest volume was at its historical low. Then a century of managed regrowth — growing stock from roughly 1.4 to 2.6 billion m^3 , +84%, most of it after the late 1960s. ► [F10b forest history](#)

⊢ REFS ✓ Aakala, Kulha & Kuuluvainen 2023, Landscape Ecol. 38:2417–2431 (“*in eastern Finland almost all productive forest land was used in slash-and-burn cultivation*”; population density the strongest predictor of depletion); ✓ Korhonen et al. 2024 (the growing-stock series)

1.4.0.2 The managed landscape still sits below its own unmanaged counterparts. Old unmanaged stands hold about 40% more carbon in the mineral soil than managed stands of comparable site types — and their own stocks do not change over a twenty-year resampling.

⊢ REFS ✓ Kumpu, Peltoniemi, Forsius & Mäkelä 2025, Silva Fennica 59:24072 (0–10 cm 2.44 vs 1.78, 10–20 cm 1.62 vs 1.14 kg C m^{-2} , both $p < 0.001$; resampled change $-692 \text{ kg C ha}^{-1}$, $p = 0.75$); □ comparability of their depth basis with our whole-profile target

1.4.0.3 ★ And the lag is long enough to matter — stated by the antecedent itself. Even with litter held at its 2004 level, Liski et al. found carbon would keep accumulating and the stocks would settle “*centuries later ... at a 38% higher level than in 1922*”. Peltoniemi et al. had already attributed part of the soil carbon of Finnish plots to a slash-and-burn history ending in the early twentieth century, and concluded that the long-term trend “*cannot be distinguished from the measured data unless the measurements cover at least two rotations*”. Three campaigns over four decades is what makes the state estimable.

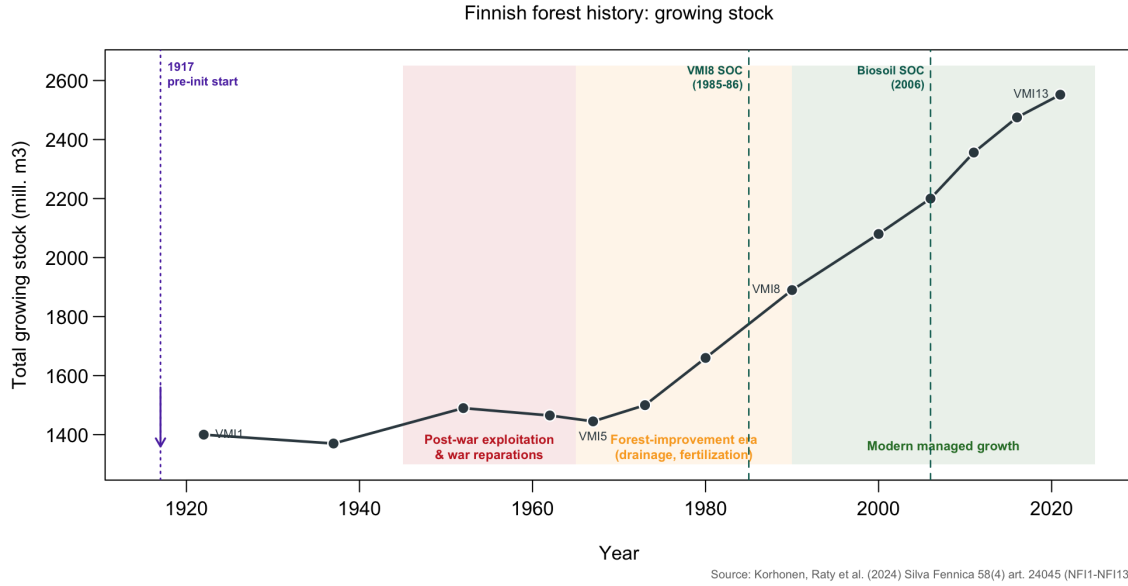


Figure 2: F10b_forest_history

⊢ REFS ✓ Liski et al. 2006 §4 (verbatim); ✓ Peltoniemi et al. 2004, GCB 10:2078–2091 (verbatim); ✓ Wutzler & Reichstein 2007 (soils disturbed *centuries* ago remain transient)

1.4.0.4 $\rightsquigarrow$ ★ And our own posteriors say the same thing, which is the reconciliation this section needs. Holding litter at its present level and letting each calibrated model run to its own equilibrium leaves a headroom of +28 to +49% above the modelled 1985 stock in five of the six — Yasso07 +44, Yasso15 +47, Yasso20 +28, TP2 +49, TP3 +82, with SP1 the outlier at +5. Liski et al. put it at +38%. **Two independent routes — their equilibrium-started forward run, our calibrated transient posterior — land on the same disequilibrium.**

⊢ REFS ours, doublechecks/equilibrium_headroom.R (arm B, 4 draws/model)

Annotation (p7): “*This need to be reconciled with our results with a discussion.*” Done, and it corroborates rather than conflicts. Note the bases differ — Liski’s 38% is relative to 1922, ours to the modelled 1985 state — so the agreement is in magnitude and direction, not decimal for decimal. SP1’s +5% is the informative exception: a single slow pool is already close to its own equilibrium, which is why it is the model that cannot produce the observed accumulation.

Consistency check with Strand 2. Liski et al. describe continued accumulation under litter frozen at 2004. Our forward arm freezes inputs at 2024 and finds the same thing — so **cite them there**, and do not present it as new. See memory **forecast-sink-robustness**.

□ **The check that would make this section decisive.** Liski et al. report soil carbon going $6.1 \rightarrow 6.3 \text{ kg C m}^{-2}$ over 1922–2004: about $+0.024 \text{ tC ha}^{-1} \text{ yr}^{-1}$, an order of magnitude below the $+0.259$ our campaigns show for 1985–2024. A model started full has little room to accumulate — the thesis of this paper, in the antecedent’s own numbers. **Verify first:** their figure is a national mean over a *changing* forest area (the denominator implies $\sim 14 \text{ Mha}$), on a “soil and litter” basis, over a different period.

1.5 Objectives

1.5.0.1 Why the method and the result are the same thing here. The convention is the reason the national trajectory has never been tested: a soil started at equilibrium reproduces a stock but has no trajectory to be wrong about, so the question of whether these models can follow four decades of change has not been asked of them. Replacing the assumption is therefore not a methodological preface to the results — it is what makes the results exist. This is stated once, here, so that a reader does not arrive at the Results expecting a methods paper.

Why this paragraph is here. The Introduction builds on the initialisation convention, which promises a *methods* contribution; the weight budget puts ~55% on the *national trend*, an empirical one. Without the joint stated explicitly, the paper reads as changing its mind at the section break.

1.5.0.2 Three questions, in the order the paper weights them. (i) Once the starting state is estimated from the measured trajectory, where does that trajectory place the models on the input/turnover exchange, relative to their published parameterisations, and how far does it narrow that exchange? (Reaching the stocks and the rate is a precondition, reported as such.) (ii) Does the position on the exchange reach the extrapolation: do two calibrations that reproduce the same window from different positions project differently, and for which structures? (iii) What is left that none of the models can see, and what would it take to see it?

Reworded 2026-09-16. (i) used to lead with “does a transient start reproduce the trajectory?” — a yes/no the Results then discount as consistency, not validation. The title says *displaces*; the question now asks *where*. Behind it sit the two advancements as Lorenzo put them: the state is *estimated* (the trajectory informs it) and its date is *pushed back* to 1917 (the error in assuming balance decays through 68 years of pre-run; what does not decay, the estimate takes up). (ii) stays on extrapolation in the Introduction; warming is named only in Strand 2.

2 Materials and methods

2.1 Data

2.1.0.1 Plots and campaigns. Finnish NFI/BioSoil/Komeetta permanent plots; three SOC campaigns (1985–95, 2006, 2024); 456 calibration-ready plots, 1205 plot-year observations.

2.1.0.2 One homogenized SOC basis. Organic plus mineral to a per-plot variable depth cap plus a modelled deep tail, reproducing the official national stocks exactly. ► [F2_baseline](#)

2.1.0.3 Litter input. The Tupek product, tree litter including harvest residues and natural mortality, excluding understorey. ► [F1_inputs](#)

2.2 The model ladder

2.2.0.1 One operational model, and three benchmarks to check it against. The object of study is Yasso — it is what the inventory runs, in three published versions. SP1, TP2 and TP3 are added as a one-, two- and three-pool benchmark ladder whose behaviour is transparent, so that any result can be read against a structure simple enough to reason about by hand. **They are a control, not competitors:** the question they answer is whether a finding is a property of the data and the transient design, or an artefact of a particular five-pool implementation.

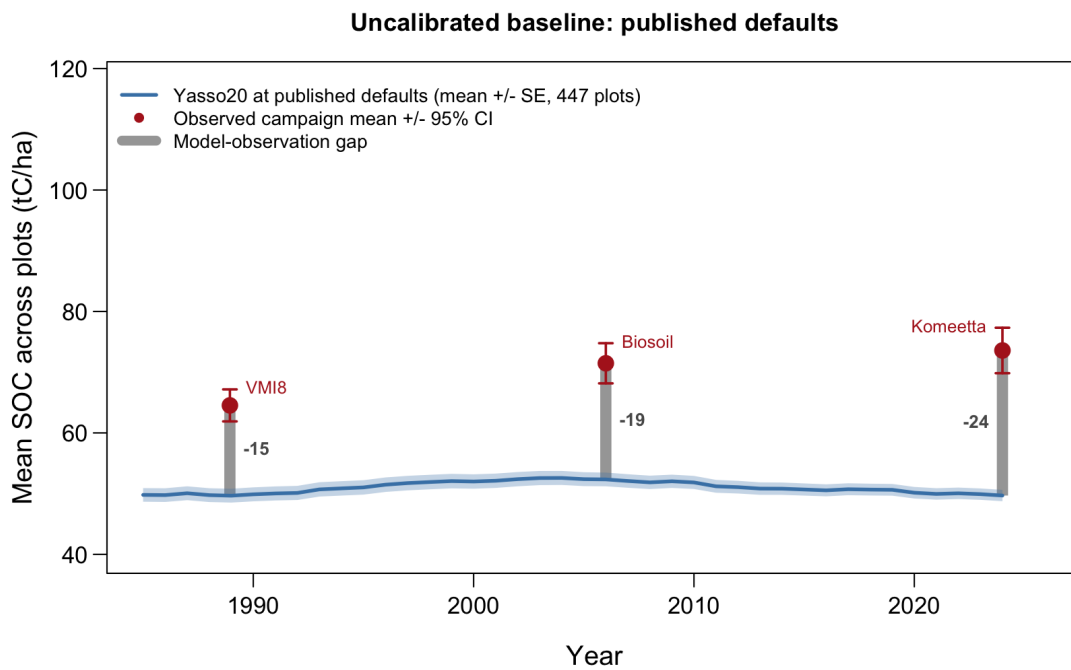


Figure 3: F2_baseline

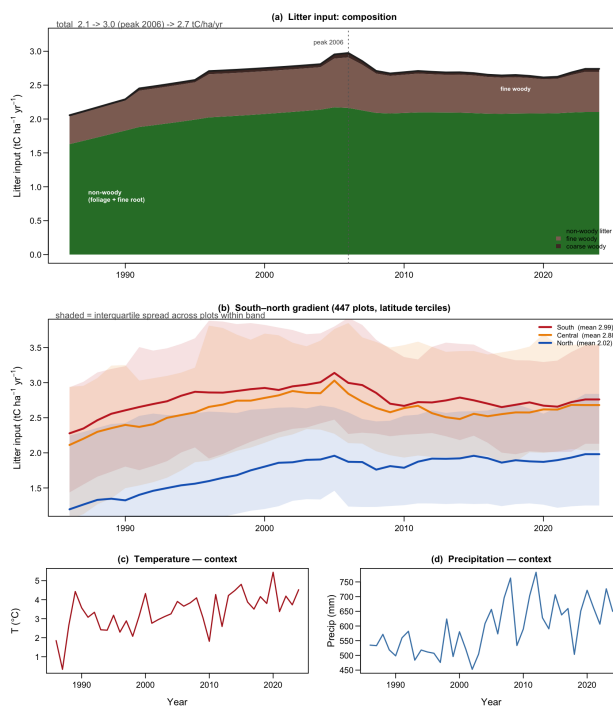


Figure 4: F1_inputs

The word “ensemble” has to be handled carefully, and this draft was not handling it. Six models run together is not an ensemble in the sense the phrase carries in this literature — a set sampled to span structural uncertainty, whose spread is a product. Ours is one operational model plus a benchmark ladder, and the spread between them is a *diagnostic*, not an uncertainty estimate. Say “all six models” where the meaning is “all of them”; keep “ensemble” for the gridded maps, where averaging the six genuinely is what is being delivered.

Annotation (p9): “we are not really doing any structural uncertainty work ... We have the simpler structures as a benchmark, mainly, with Yasso as the proven standard.” **Adopted:** the ladder is described as a benchmark design, and the word “structural uncertainty” is used only where it is earned (the forward spread), never as a product claim.

2.3 Transient initialisation

2.3.0.1 The start becomes a parameter. Each draw spins up from 1917 under a reconstructed input ramp and is carried forward through all three campaigns; σ_{init} scales the 1917 input level. ► [F4_initialization](#)

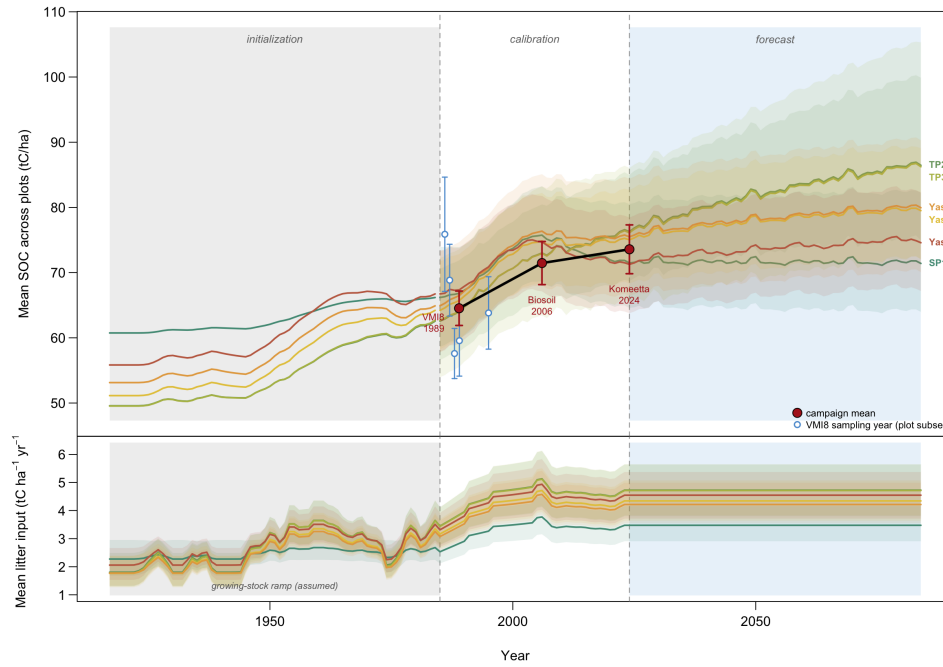


Figure 5: F4_initialization

2.4 A fair frame

2.4.0.1 Equal information, not equal parameter count. Yasso does not fit its decomposition rates, so the simple models do not either: intrinsic rates are externally anchored (ICBM), humification fractions are free, and every model calibrates the same auxiliary pair.

2.5 Calibration and error model

2.5.0.1 One likelihood, and it does not pretend plots are independent. Log-normal, with region, plot and campaign offsets marginalised at fixed variances.

2.6 Analyses

2.6.0.1 Three, matching the three objectives. Posterior predictive over the observed window; a forward experiment contrasting published and calibrated parameters under climate scenarios; and a residual analysis against site covariates.

3 Results and discussion

Combined, deliberately — and nested at the SUBSECTION level. Every result here needs its interpretation immediately: the level agreement means nothing without the ridge, and the ridge means nothing without what it costs. **The rhythm inside each subsection is fixed:** one paragraph states what was measured and stops; the next, marked $\rightsquigarrow$, interprets it and opens with an explicit signal (“*This implies*”, “*Two readings*”, “*So*”). A result paragraph never takes an inference in its stress position. **Why not nest at paragraph level:** the reader loses track of which sentences are observation and which are inference, and the split stops being mechanical — `\combinedfalse` would then mean unpicking every block by hand instead of moving whole paragraphs. Cross-cutting argument that no single result owns (the ridge’s meaning, the limitations) stays in a short closing Discussion.

STRAND 1 — THE NATIONAL TRAJECTORY ($\sim 55\%$ of the paper)

3.0.1 The models reach the observed levels

3.0.1.1 No model sits more than 3.5 tC ha^{-1} from any campaign mean. On the current calibration all six models reproduce all three campaign levels. [► F3_mean_soc](#)

3.0.1.2 $\rightsquigarrow$ Reaching those levels is not the free parameter it would be in a conventional calibration. The equilibrium stock is $J \times \tau$, and if both factors were free, hitting a level would be arithmetic rather than a result. Neither is free here. The decomposition rates are externally anchored and not fitted — Yasso’s a -vector is a fixed constant, the simple models’ fast rate likewise and their slow rate pinned at the ICBM value — and the input multiplier is held to a prior of log-SD 0.25 around an independently derived centre. What the six models are asked to do is reach *three* campaign levels, decades apart, on anchored kinetics and a constrained input, from a start that is itself inferred. That they all do is the evidence that the fair frame works.

3.0.1.3 $\rightsquigarrow$ What it cannot do is discriminate. Every point on the input/turnover ridge reaches the same level by construction, so agreement here separates neither the six structures from one another nor our calibration from the published one. It is the precondition for the two claims that follow — the rate and the shape — and it is read as such, not as support for them.

Annotation (p7): “I don’t get what this refers to.” The v2 text pointed at a “ 26 tC ha^{-1} too high at 1985” caveat that no reader could know about — it was an internal memory of an earlier run. **Deleted;** the sentence now states the present result only.

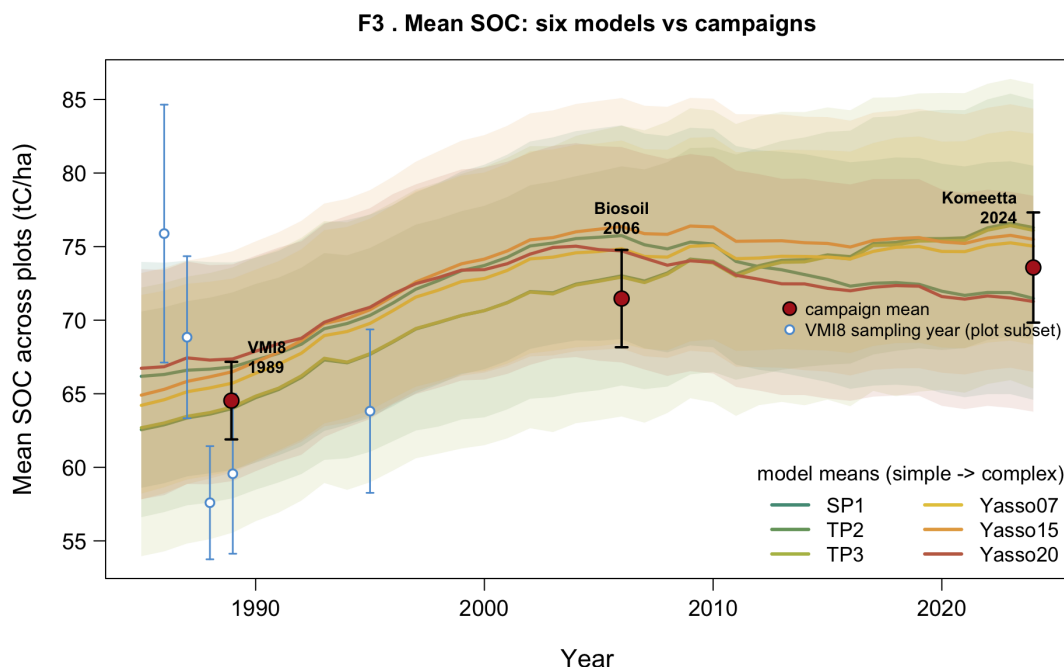


Figure 6: F3_mean_soc

3.0.2 And they reproduce the rate, within the limits of the data

3.0.2.1 The six models bracket the observed rate, and two land on it. Yasso07 +0.258 and Yasso15 +0.250 against an observed +0.259; TP2 and TP3 run stronger, SP1 and Yasso20 weaker.

► F5_stock_change

3.0.2.2 ~ The bracket is the result; the two that land on it are not. That the six span the observed rate says the transient design produces the right order of magnitude of accumulation, which an equilibrium start cannot. Which members happen to fall closest is not interpretable: they were not selected for it, the observed rate carries its own comparability error, and the spread between models is a diagnostic rather than an uncertainty.

Stated as consistency, not as validation (annotation, p7). The observed rate carries its own comparability error, so “two models land on it” is reported as what was seen, with the campaign caveats attached, and no model is preferred on this basis.

3.0.3 The trajectory shape splits the six models three ways

3.0.3.1 Three behaviours, and they are legible. Yasso07 and Yasso15 rise and flatten; TP2 and TP3 rise and keep rising; SP1 and Yasso20 peak in the late 2000s and decline. The same split is visible in the per-plot deliverable. ★ TO BUILD: F3b — trajectory shape, three groups, explicitly grouped

Annotation (p7): “Try it, but then we have to look at it together before including; current Fig 6 in this text could be enough already.” Kept as TO BUILD — and note that the grouping is already visible without a new figure in the NextGenC trajectory panel, which may make F3b unnecessary. Decide when it exists.

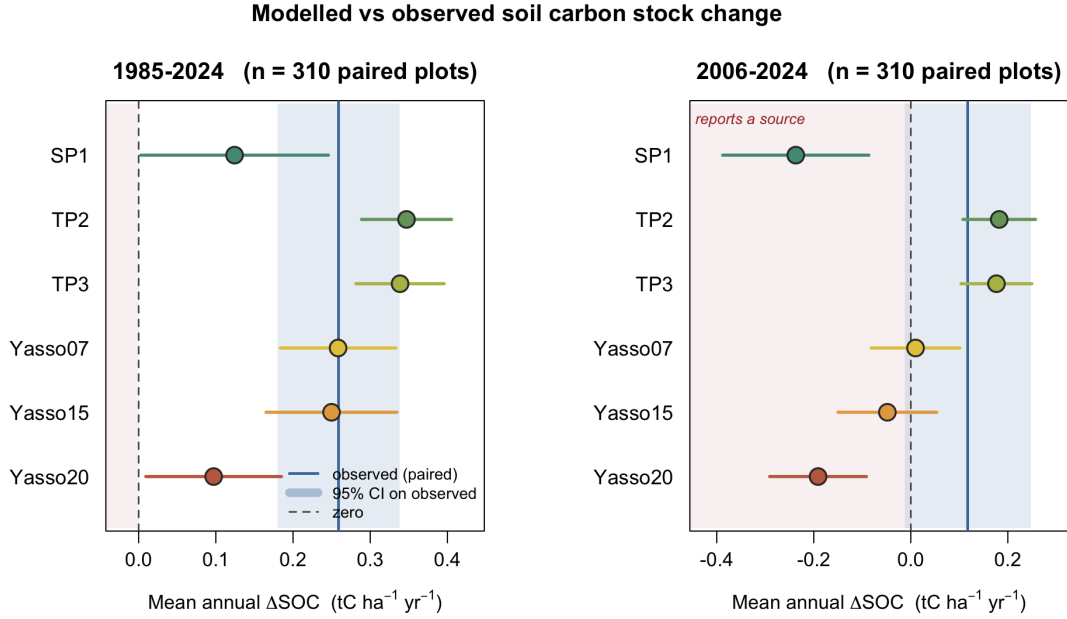


Figure 7: F5_stock_change

3.0.4 This is what an equilibrium start forecloses

3.0.4.1 The shape is the point. Accumulation bending over is a statement about timing, and timing is precisely what an equilibrium initialisation discards. Reproducing it is the paper's first result.

3.0.5 Structure is nearly invisible in fit

3.0.5.1 Skill does not climb the pool ladder. Calibration R^2 0.016–0.021; holdout $R^2 \leq 0.0012$; holdout RMSE spans 3.6%, and the best and worst belong to a five-pool and a three-pool model. [► T1_model_summary](#) [► F6b_skill_vs_complexity](#)

Model	Pools	Free params	R^2 (cal)	R^2 (hold)	RMSE (hold)	Eff. flux $\sigma_{\text{inp}} \bar{J}$	σ_{init}
SP1	1	6	0.017	0.000	47	3.2	0.84
TP2	2	7	0.016	0.000	48	4.3	0.50
TP3	3	9	0.015	0.000	49	4.3	0.49
Yasso07	5	20	0.019	0.000	48	4.0	0.52
Yasso15	5	26	0.021	0.001	47	3.9	0.54
Yasso20	5	26	0.016	0.000	47	4.2	0.59

3.0.5.2 $\rightsquigarrow$ So fit cannot arbitrate between these structures. Whatever separates them, it is not the calibration data — which is why the benchmark ladder is read as a control on the operational model rather than as a competition, and why Strand 2 has to ask the question somewhere else: in extrapolation, where the structures do separate.

3.0.6 And plot by plot there is no gradient to fit

3.0.6.1 The models impose an ordering across plots; the observations do not have one. Holdout R^2 is zero to three decimals in all six, and the fitted line through predicted-against-observed

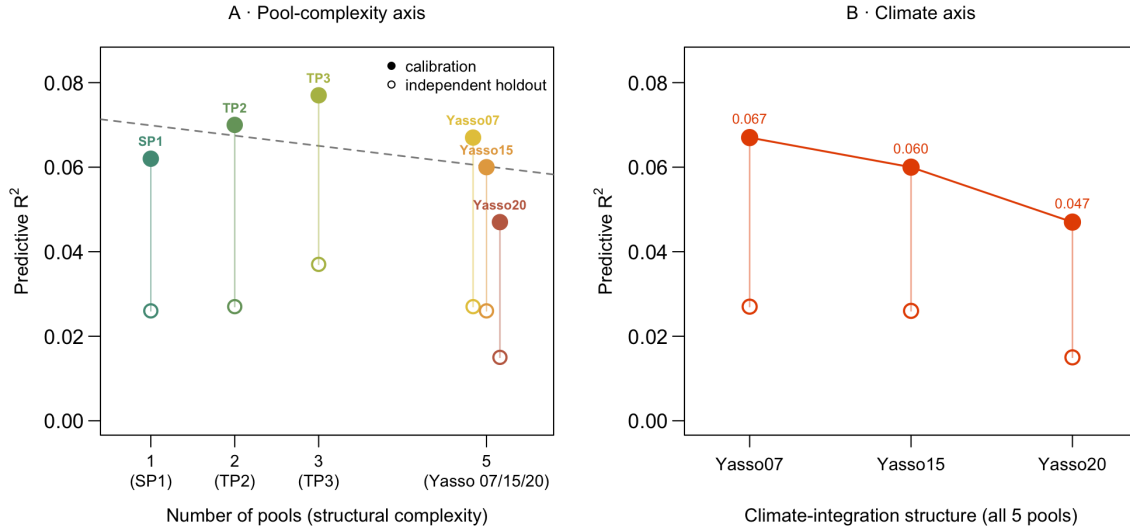


Figure 8: F6b_skill_vs_complexity

is flat. But the informative part of the figure is not the flatness — it is that the *predicted* spread is the wider of the two (0–270 against an observed 30–220), and that it is cleanly ordered by stand basal area, the models’ only real between-plot driver, while the observations show no such ordering.

► F6_obs_vs_pred_holdout

3.0.6.2 ~ So the problem is not that the models are uninformative about plots — it is that they are confidently wrong about them. Litter input varies across plots and the models translate that faithfully into carbon; observed SOC is ordered by something else. Between-plot variation in this landscape is dominated by local factors — edaphic, hydrological, historical — for which no model here has a term, and against which a litter-driven gradient is not merely weak but misdirected.

3.0.6.3 ~ An ordering the models do not carry is a question, and it is the last strand’s. If the observations are ordered by something absent from every model here, the question is what — and that question is answerable with data already in hand (Strand 3).

3.0.6.4 ★ Measured: the models express a between-plot gradient an order of magnitude steeper than the soil’s. Regressing log SOC on stand basal area gives a model response of +0.044 to +0.049 per $\text{m}^2 \text{ha}^{-1}$ ($r = 0.62\text{--}0.73$) against an observed +0.0044 ($r = 0.11$). The residual therefore falls with basal area in **all six models**, at -0.032 to -0.044 , with p from 10^{-16} to 10^{-111} and on holdout, calibration and pooled data alike. Across the observed range of $40 \text{ m}^2 \text{ha}^{-1}$ the models make soil carbon span a factor of **6.3**; the observations span **1.19**.

└ REFS ours, doublechecks/basal_area_sign.R (arm B run; residual basis log obs – log pred, identical to the RF target)

3.0.6.5 ~ And the little that the data do show is mostly geography. Controlling for region and soil type, the observed slope falls from 0.00438 to 0.00184 and stops being significant ($p = 0.12$), while the model slope is unmoved at 0.0444. On that adjusted basis the mismatch is a factor of **24**, and the observed span across the whole basal-area range is 1.08 — statistically indistinguishable from no dependence at all.

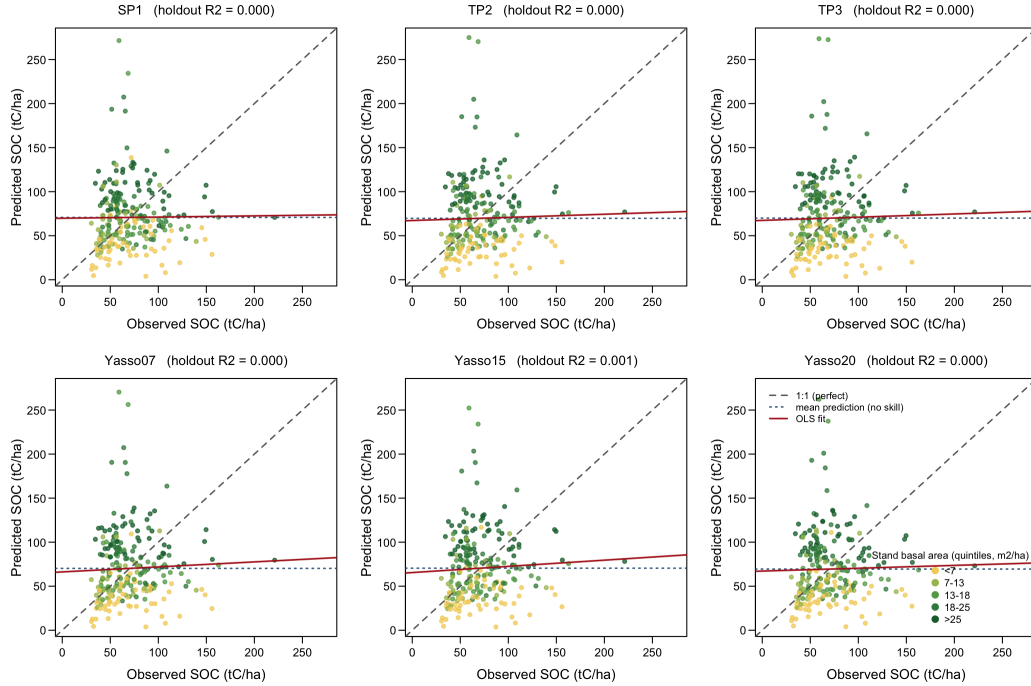


Figure 9: F6_obs_vs_pred_holdout

3.0.6.6 Where the gradient comes from: the litter product, transmitted faithfully. The models carry no basal-area term. They carry litter input, and matched in time to the same 1985 basal area that gradient is $d \log J / dBA = +0.072$ — a span of $\times 18$ across the observed range. The modelled stock's own slope is $+0.073$ to $+0.076$. **The models pass the input gradient through essentially one-for-one**, and the forward run then damps it (the predictive slope is $+0.044$ to $+0.049$). There is no amplification to explain and no defect to find.

└ REFS ours, doublechecks/basal_area_sign.R and woody_compartment_test.R

✓ **Tested and refuted 2026-09-04: it is not a compartment mismatch.** The hypothesis was that Yasso's stock includes woody debris a soil core cannot contain, so the mismatch would grow with standing biomass. Zeroing the woody litter inputs moves the gradient by 6% (Yasso15 $0.0755 \rightarrow 0.0710$) against a $15\times$ gap to the observations. **And the decisive control is free:** SP1, TP2 and TP3 carry a single scalar litter flux with *no* composition information at all, and they produce the *same* slope (0.073). A mechanism absent from three of the six cannot explain a gradient present in all six.

3.0.6.7 ~ So the finding belongs to the input product and to the soil, not to the models. The litter product says a plot's carbon input varies eighteen-fold across the basal-area range; the soil says its carbon varies by a factor of 1.2. Two things follow, and the second is the more useful. (i) Plot-level soil carbon is not set by current litter input — each plot sits at a different point on its own disturbance trajectory, and one global initial state cannot express that. (ii) **An eighteen-fold input gradient is not credible on its face.** A stand-keyed product gives a freshly cut plot almost no litter, and a freshly cut plot does not stop receiving carbon: ground vegetation, dying roots and residues continue. The plot-scale failure is therefore evidence for the *same* missing flux as §3.0.9, reached from the opposite end — and it points at the two components that section is weakest on, understorey and early-succession vegetation, because those are exactly where a low-basal-area plot's input has gone.

3.0.7 The data pin a level, and little else

3.0.7.1 Convergence is not identifiability. R-hat ≤ 1.008 everywhere, yet the transfer fractions are prior-pinned, ridge-correlated and carry under a nat. ► [F8b_fraction_ridges](#)

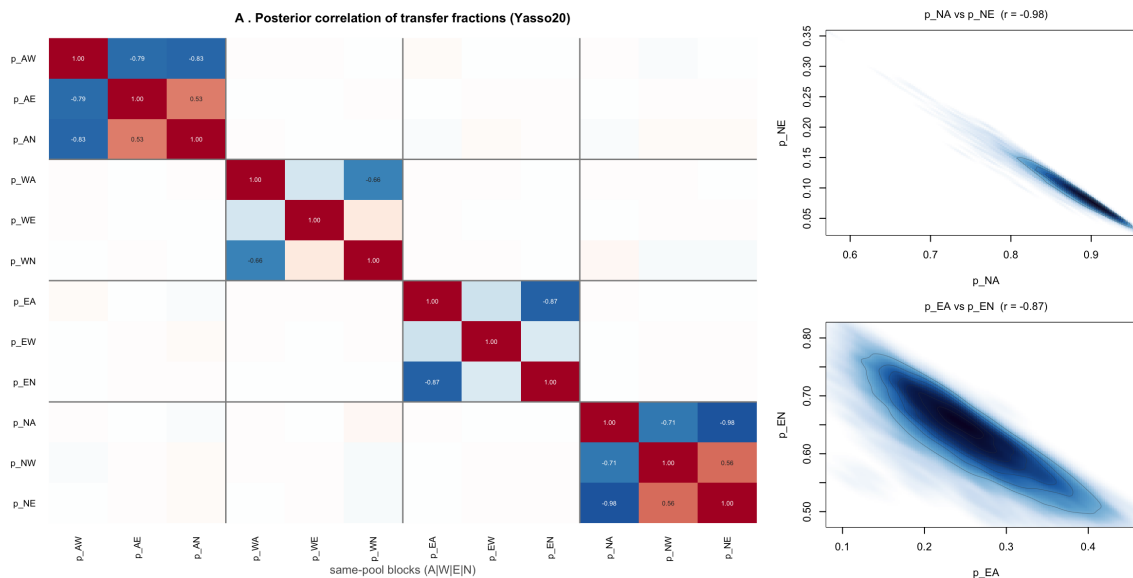


Figure 10: F8b_fraction_ridges

3.0.7.2 Clean convergence on a non-identified posterior is the expected result of fixing the mixing, not a failure to notice a problem. Clean R-hat with prior-pinned marginals, near-perfect anti-correlation between fractions leaving the same pool, and sub-nat information gain are three independent signatures of the same thing: the data constrain the net flux out of a pool, never the split. It also sets the limit of what follows — what the campaigns identify is a level and a rate, which is what the inventory needs and is all the paper claims.

3.0.8 What fitting the trajectory requires: more input, and faster turnover

3.0.8.1 The calibration lands away from the published parameterisations. Effective litter flux 3.19–4.34 tC ha⁻¹ yr⁻¹, and intrinsic MRT 24.6 / 27.1 / 22.6 yr for Yasso07/15/20 against published 33.5 / 30.5 / 22.1. ► [F9_auxiliary_sigmas](#) ► [F12_mrt_yasso](#)

3.0.8.2 The input displacement is the visual hook, and it is a biological statement. Every model settles above the tree-only litter product it was given ($\bar{J} \approx 2.5$) and inside the region-matched NPP envelope — so what the calibration asks for is not physically impossible, it is a flux the inventory product does not contain. That plate carries the paired auxiliary result: the same posteriors start 1917 well below equilibrium. It is the figure this subsection opens on; F12 and F14 then say what the input displacement costs in turnover.

Restored 2026-09-04. F9 was built and current but referenced nowhere in v3 — the envelope plate is the one picture that makes “more input than the product supplies” visible, so it belongs here rather than only implicitly behind the ridge. **Direction check:** relative to *published* our turnover is **faster**, not slower (24.6/27.1 vs 33.5/30.5); it is relative to *earlier runs of ours* that MRT has lengthened. The ridge makes more input pair with faster turnover, and the paragraph title says so.

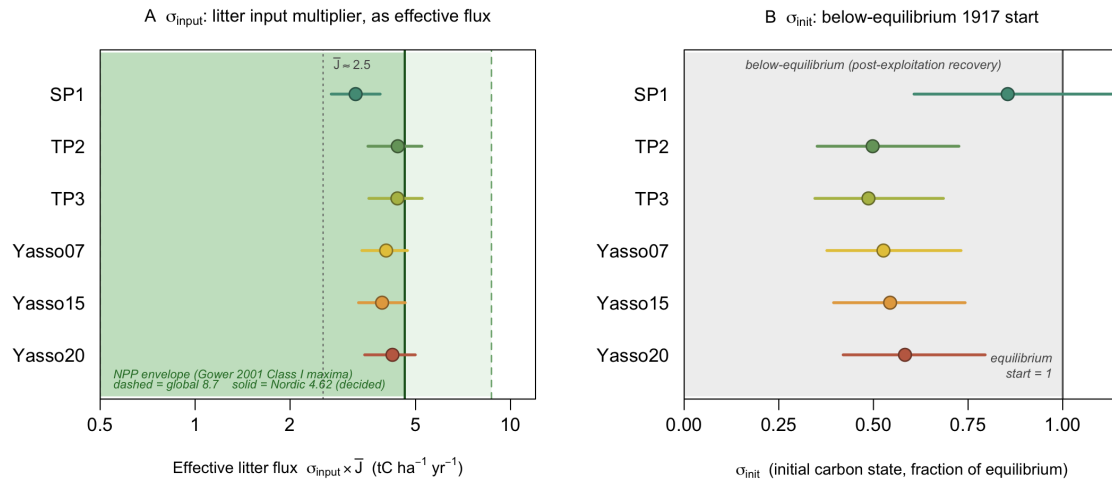


Figure 11: F9_auxiliary_sigmas

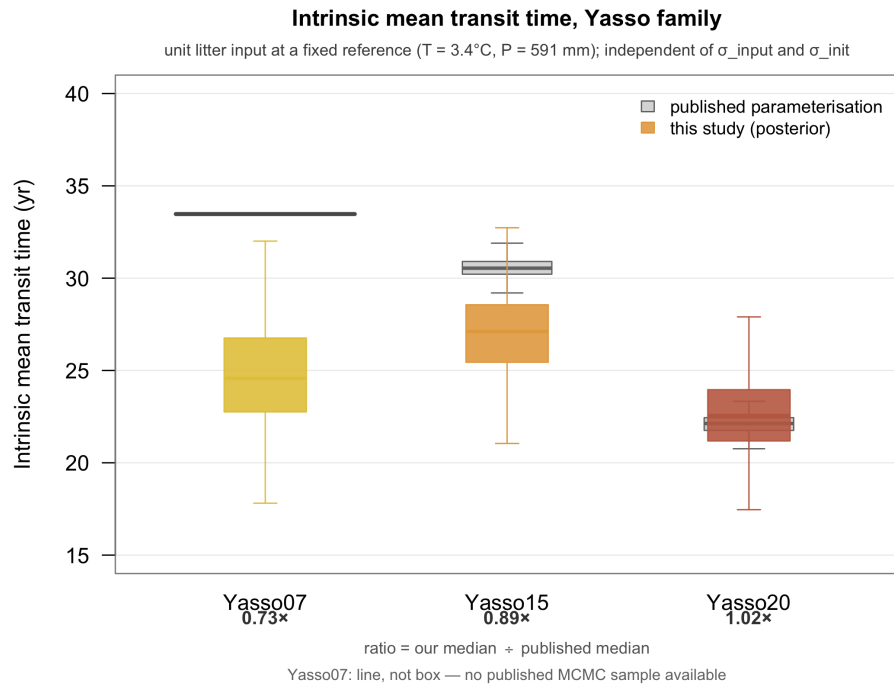


Figure 12: F12_mrt_yasso

3.0.8.3 $\rightsquigarrow$ These are one displacement, not two. The level constrains $MRT \times \sigma_{\text{input}}$, so faster-with-more and slower-with-less fit alike; the posterior rides that ridge. $\blacktriangleright$ F14_mrt_ridge $\blacktriangleright$

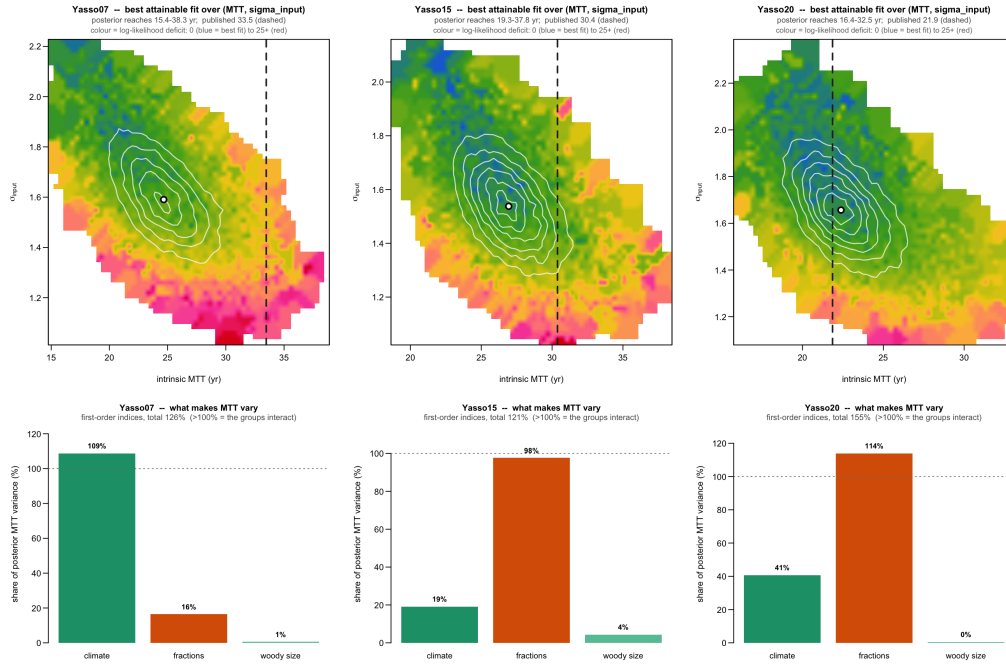


Figure 13: F14_mrt_ridge

S13_mrt_ridge_benchmark

Annotation (p12), on “a steady-state start hides exactly this”: “Steady state still has the same problem of equifinality ... with the trajectories the ridge on the likelihood surface should narrow a bit ... no need to overdo this surprise thing.” **Rewritten.** The claim is now the weaker and defensible one: the ridge exists under either initialisation; a transient start narrows it, because the approach rate carries information the equilibrium condition discards. Measured, the narrowing is partial — MRT and σ_{input} still correlate -0.64 to -0.73 . The “surprise” framing is gone; the displacement is reported as a result, once.

Annotation (p12), on Liski: “I don’t think so, we are correcting even higher.” Confirmed and corrected. Liski 2006 implies $\sigma_{\text{input}} \approx 1.27$; our effective correction runs 1.27 – 1.73 across the six. Liski is the floor of our range, not a match — so the sentence becomes “an independent estimate from the same country, period and model family sits at the bottom of the range we infer”, which is a weaker corroboration and an honest one.

3.0.9 What might be missing from the input flux

Annotation (p12), the longest one: “The input functions have a lot of trust in the field ... they do work (I think because the previous fitting compensates) ... there needs to be a very extensive review of the supporting evidence, with hypotheses well grounded in literature about what we might be missing in the fluxes (first things to examine are fine root MRT , exudates and mycorrhizae) ... also missing pioneer vegetation.” This section is the answer to that note, and it is the main place where the paper stops being about models.

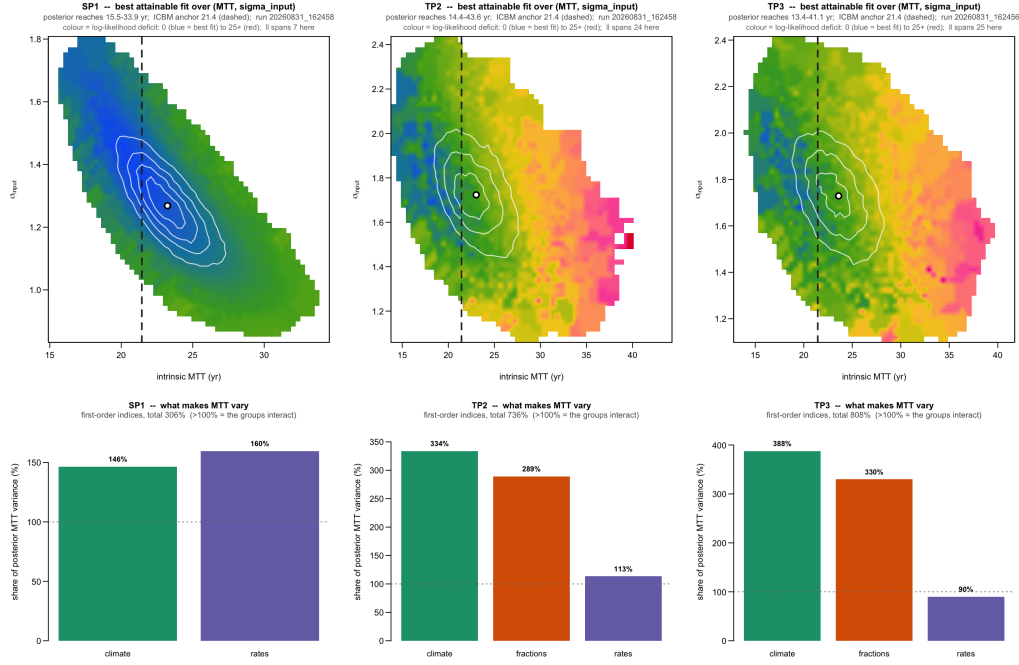


Figure 14: S13_mrt_ridge_benchmark

3.0.9.1 The candidate is not that the litter model is wrong, but that it is incomplete. Litter input products are trusted because they have worked for decades — and one reason they work is that a calibration free to adjust rates can absorb a missing flux without visible misfit. A transient calibration on a fixed rate anchor removes that compensation, and what is missing then has to appear somewhere.

3.0.9.2 The size of what has to be explained. Effective flux 3.19–4.34 against a supplied $\bar{J} = 2.511 \text{ tCh}^{-1} \text{ yr}^{-1}$: a gap of **0.68–1.83**. The candidates below are ordered by how well the literature prices them, *not* by how well they help.

└ REFS ours, arm B run 20260831_1624*; ✓ Liski et al. 2006 implies $\sigma_{\text{input}} \approx 1.27$ independently — the floor of our range

3.0.9.3 Rhizodeposition — the one with an accounting precedent. Cropland carbon accounting has carried this term for two decades: the standard approach adds an “extra-root material” pool at 65% of standing root biomass. Forest inventory chains add nothing analogous. And root-derived carbon is not a low-efficiency input — at Ultuna its optimised humification coefficient is about $2.3\times$ that of the same mass of aboveground residues.

└ REFS ✓ Bolinder et al. 2007, Agric. Ecosyst. Environ. 118:29–42 (the $\times 1.65$; RD coefficients in the literature span 0.5–2.0); ✓ Kätterer, Bolinder, Andrén, Kirchmann & Menichetti 2011, AEE 141:184–192 (the $2.3\times$); □ a boreal-forest rhizodeposition estimate

The cropland factor cannot be carried over at face value. In cropland, root biomass is a harvest-time *snapshot*, so the 65% compensates for within-season turnover *and* root hairs *and* exudates. A forest chain already counts fine-root turnover explicitly (at 0.85 yr^{-1}), so the forest analogue covers only what is left and **must be smaller**. The claim is therefore: cropland accounting recognises a belowground input beyond root biomass and prices it; forest accounting recognises none; the truth is in between. ✓ **Answered 2026-09-04, and it weakens this objection.** Bolinder et al. derive the 0.65 as $33/50$ — the share of belowground C *released by living roots and remaining in soil* over the share *remaining in roots* — and state that the extra-root pool is “*exudates, as well as root hairs and fine roots sloughed off during the growing season which, because of sampling difficulties, are not included in the ‘root’ fraction.*” That is not the within-season turnover a forest chain already counts; and forest fine-root biomass is estimated from cores and root:leaf allometry, which has the *same* sampling problem. □ a boreal tracer estimate of the same ratio; □ whether our chain’s fine-root biomass misses sloughed material.

3.0.9.4 Understorey — outside the litter product by construction, though not outside the inventory. The national inventory carries a ground-vegetation litter term; the litter product used here is a tree-litter product and does not, by design. The understorey share also rises strongly northward, so a single national multiplier is wrong in opposite directions at the two ends of the country. ► [SX_input_vs_literature](#)

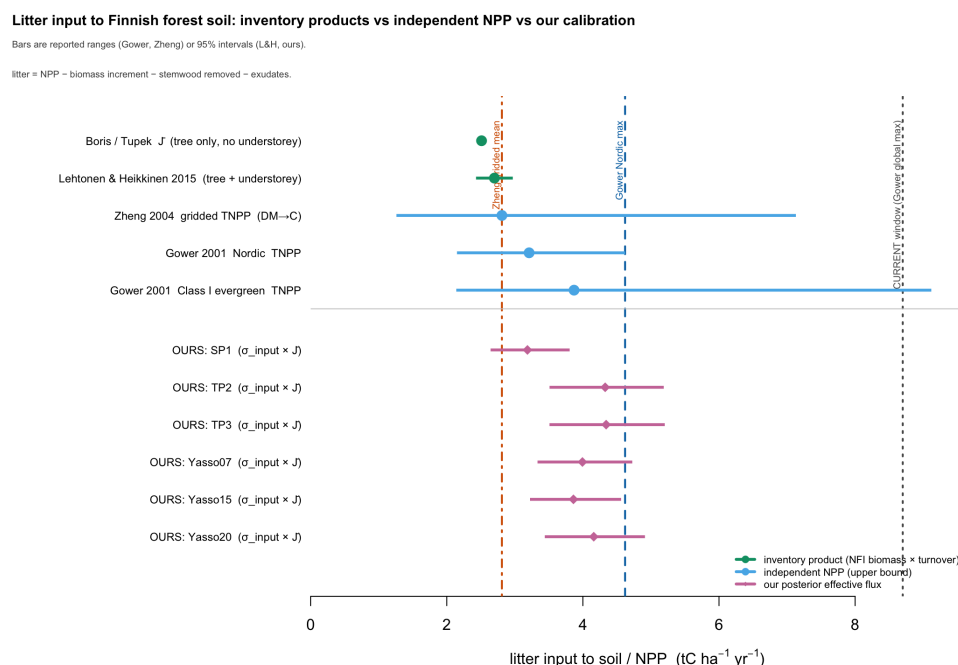


Figure 15: SX_input_vs_literature

Annotation (p19): “Of course this formulation cannot be in the manuscript...” Agreed and rewritten. “The product we were given” reads as a complaint about a collaborator. The product excludes understorey *by design*, which is a statement about scope, not a shortcoming — and Boris is a coauthor. Phrase it as scope everywhere.

└ REFS ✓ NID 2024 Table 6.4-3 via Muukkonen et al. 2006 ($50.6 \text{ S} / 66.6 \text{ N gC m}^{-2} \text{ yr}^{-1}$ aboveground); ✓ Lehtonen et al. 2016 (“*the role of understorey litter input was underestimated ... especially in northern Finland*”); ✓ Lehtonen & Heikkinen 2015 (our anchor)

DECIDED (Lorenzo, annotation p20): stay on arm B for now. “For the moment I would stick just to Arm B, and go with it, we see later and eventually when referee revise it.” No re-run on the anchor. The arithmetic below is recorded because it must be *stated* in the Methods, not because it changes the run.

★ **Resolved 2026-09-04 — the sources agree; our derivation did not.** Lehtonen & Heikkinen carry the *same* understorey litter as the inventory, in their own Table A5 (506 S / 666 N kg C ha⁻¹ yr⁻¹, from Muukkonen & Mäkipää 2006), and Liski’s implied 0.61 sits inside that range. The 1.08 we ran came from dividing L&H’s *absolute total* (2.70) by *our* $\bar{J}$ (2.511) — two different litter products, so it measures the understorey gap *minus* the offset between their tree-litter levels. Adding an understorey estimate to our own $\bar{J}$, on our own plot distribution, gives **1.21**; Liski gives **1.27**. **Arm A and arm B were never a sensitivity pair.** □ provenance of the 2.70; □ region coding. Owner: Aleks Lehtonen.

3.0.9.5 Mycorrhizal necromass. Root-associated fungi are implicated as a dominant precursor of stable boreal soil carbon, and mycelium is not a tree biomass compartment, so no turnover coefficient can produce it.

└ REFS ✓ Clemmensen et al. 2013, Science 339:1615–1618 (50–70% of stored carbon from roots and root-associated organisms); □ Ekblad et al. 2013 and Hagenbo et al. 2017 (production, in carbon); □ the necromass fraction of production

Do not quote the 50–70% as the size of a missing flux. It describes the *origin of stored carbon*, and some of that root-derived carbon is already in our input as fine-root litter. The defensible sentence: the dominant precursor is root-derived and partly fungal, and *the fungal component* is outside inventory litter models.

3.0.9.6 Fine-root turnover — and here the evidence points the other way. Minirhizotrons put Norway spruce fine-root lifespan below 2.5 yr; radiocarbon returns apparent ages up to 14. A reconciled mean residence time of ~ 3.8 yr would imply a turnover near 0.26 yr⁻¹ against the 0.85 in use — which would make fine-root litter *smaller* and **widen** the gap.

└ REFS ✓ Strand et al. 2008, Science (the disagreement); □ Hansson et al., New Phytol. (60% within 0.75 yr, remainder within 8.4 yr — **blocking**: a flux is carried by the fast members, so check whether 0.85 is a defensible flux-weighted value); □ provenance of the 0.85 in our own chain — one email to Boris or Aleks

3.0.9.7 Early-succession vegetation. Litter models keyed to standing tree biomass are least informative exactly where that key is smallest: the years after a stand-replacing harvest.

└ REFS □ Palviainen et al. (ground vegetation after clear-cutting; biomass falls 46–65% first, the pioneer flush comes later — **the sign over a rotation is not established**); □ **our own residuals against time since disturbance** — stronger than any citation here, and a few hours of work

This section must not become an argument that we are right. Two disciplines enforce it. (i) One candidate points the wrong way (fine roots) and it stays in. (ii) The priced candidates now *overshoot*: understorey + mycorrhizal + rhizodeposition at full cropland weight is 1.4–2.1 against a gap of 0.68–1.83. Present them as a set that **brackets** the gap, never as a sum that matches it — a budget adding up exactly would be the more suspicious result. The honest form remains: the calibration says the input is larger than the product; here are physically plausible sources; testing them is not this paper’s job.

STRAND 2 — DOES THE STARTING-STATE ASSUMPTION REACH THE WARMING RESPONSE? ($\sim 25\%$)

Annotation (p17): “For what? Do you mean if it has consequences for extrapolation, for example?” Yes — and the section is now titled that way. v2’s “does the ridge position matter?” left the reader to guess the predicate. It matters *for extrapolation*: two calibrations that fit the observed window identically can project differently.

The link that was missing (2026-09-16). The strand used to ask whether ridge position matters, without saying *why we are at this position*. We are here because the start was estimated: all six models put the product $J \times \tau$ 28–49% above the observed stock (the headroom), whereas a balanced start sets that product *equal* to the stock. A balanced start with free rates therefore lands at a smaller product — faster turnover, or less input — and stores the missing history in the rates (Wutzler & Reichstein 2007). In a single-modifier structure the rates *are* the temperature sensitivity, so the initialisation assumption reaches the warming response through them. Inductively, the direction for Yasso07 is a *steeper* response than ours, and ours ($\xi \times 2.13$, β_1 0.099 $\rightarrow$ 0.158) is already steeper than published. **Tested:** two positions on the ridge, F15. **Not tested:** the balanced-start arm itself (NEXT_RUN_equilibrium_init.md) — the initialisation $\rightarrow$ position step is deductive.

3.0.10 The question the ridge forces

3.0.10.1 The data cannot choose a point on the ridge, so the choice must be judged by its consequences. If moving along it changes nothing we can observe, the displacement is book-keeping. If it changes the projection, it is a result with consequences for anyone using these models.

3.0.11 The test: published parameters against ours, forward only

3.0.11.1 No re-calibration, one input parameter refitted. Both arms are pushed through the same machinery under ramped scenarios; both match the observed window at least as well as each other, so any divergence afterwards is extrapolation. ► [F15_forward_arms](#)

3.0.12 It matters for one structure and not the others

3.0.12.1 Only Yasso07 separates. $-2.08 [-3.57, -0.79]$ tC ha $^{-1}$ at SSP2-4.5; Yasso15 and Yasso20 do not separate at all.

3.0.12.2 $\rightsquigarrow$ And the mechanism is structural. Yasso07 applies a single climate modifier to every pool, so turnover and temperature sensitivity are tied to the same parameter and moving along the ridge moves both. Yasso15 and Yasso20 carry pool-specific modifiers, and the pools that *hold* the carbon respond almost identically in both arms. ► [A1_forward_appendix](#)

3.0.12.3 $\rightsquigarrow$ The framing that survives. Yasso15 and Yasso20 are not shown to be right; they are shown to be **resilient** to an input assumption the data cannot settle. That is a usable statement for an inventory, and a narrower one than “our calibration changes the forecast”.

The main forecast contains no climate change, and the paper must say so. The projection recycles the last twenty observed years and freezes litter at 2024, so its sink is relaxation toward equilibrium, not a response. The scenarios appear only in this side arc, as an instrument for testing ridge position. Doing it properly would need productivity to respond to warming as well.

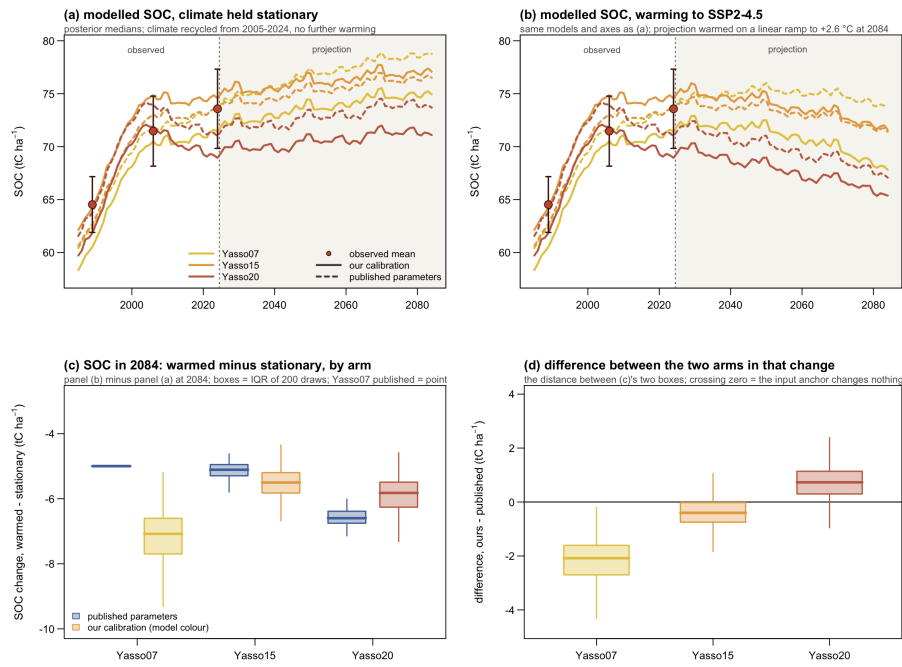


Figure 16: F15_forward_arms

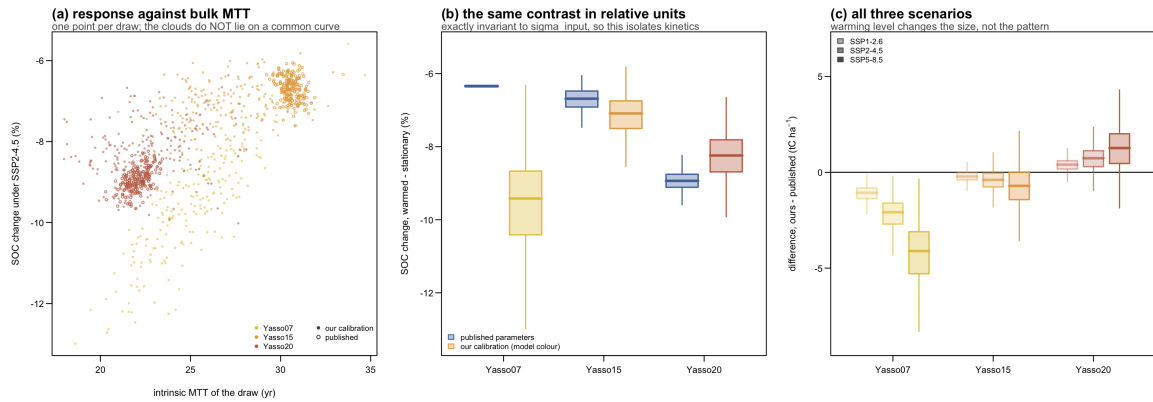


Figure 17: A1_forward_appendix

STRAND 3 — WHAT NONE OF THE MODELS CAN SEE ($\sim 20\%$, and the last part brief)

Annotations (p16, two): “*This seems about local variance, point variance, rather than the overall trend ... This is a separated topic. I would put it lower in priority ... we are just offering this analysis to raise new research questions ... Our main story is about the global, national variance.*” **Restructured accordingly:** this is no longer a subsection inside the input argument (where v2 put it, and where it did read as evidence for the input claim). It is its own strand, last, at $\sim 20\%$, and it is framed as a question raised rather than a gap filled.

3.0.13 The models order the national trajectory but not the plots

3.0.13.1 Two statements that read as a contradiction only if separated. All six models reproduce the national mean trajectory and explain almost none of the between-plot variance. Both are true, and the second is a statement about spatial resolution, not about the trajectory.

3.0.14 What is left behind follows the history of each stand [rebuilt 2026-10-07]

Lorenzo, 2026-10-07: two key figures, in this order, with the reconciliation between them — “two sides of the same story”. The basal-area effect first, then its decomposition by time since the cut, the connection, and a pointer to **pioneer vegetation** (a known gap, much ongoing modelling work: Mehtätalo at LUKE, Swedish groups); the site-ecology parts are minor but interesting. How far to develop the pioneer point is to be discussed. **F11 (the RF heatmap) opens the strand:** basal area emerges from it, and the rest was developed from its leads.

3.0.14.1 The entry point — what the random forest finds. About half of the residual is recoverable: 44–65% out of bag per model (observation repeatability ≈ 0.49). The same leads in all six: **basal area 1985** by far, then development class 1985 and coarse fragments. ► [F11_rf_heatmap](#)

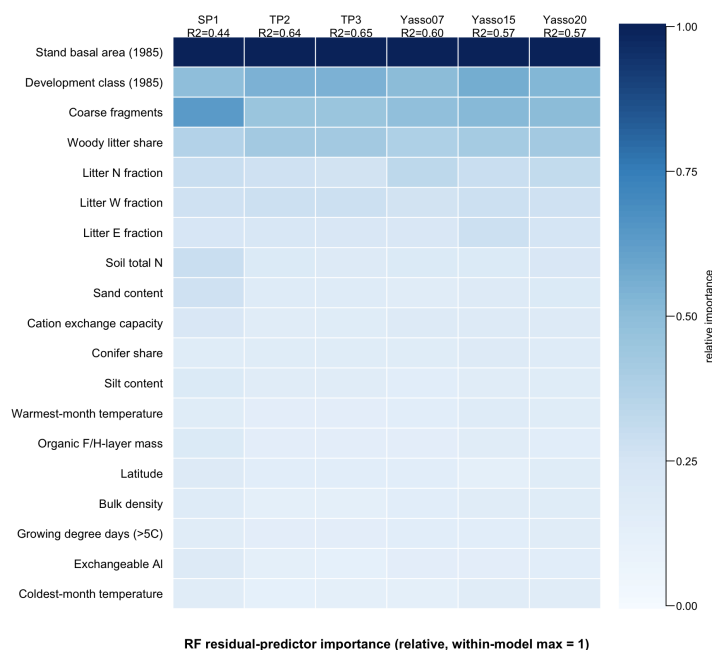


Figure 18: F11_rf_heatmap

The detailed analysis follows those leads; a wider set of several hundred variables predicts no better (56%) and confirms the ranking.

3.0.14.2 Four variables carry almost all of it. A regression on 1985 basal area plus fertility, texture and stoniness explains 53%. Residuals of the six models correlate ≥ 0.99 , so the mean of Yasso07 and Yasso15 stands for all. ► [F11_stand_and_soil](#)

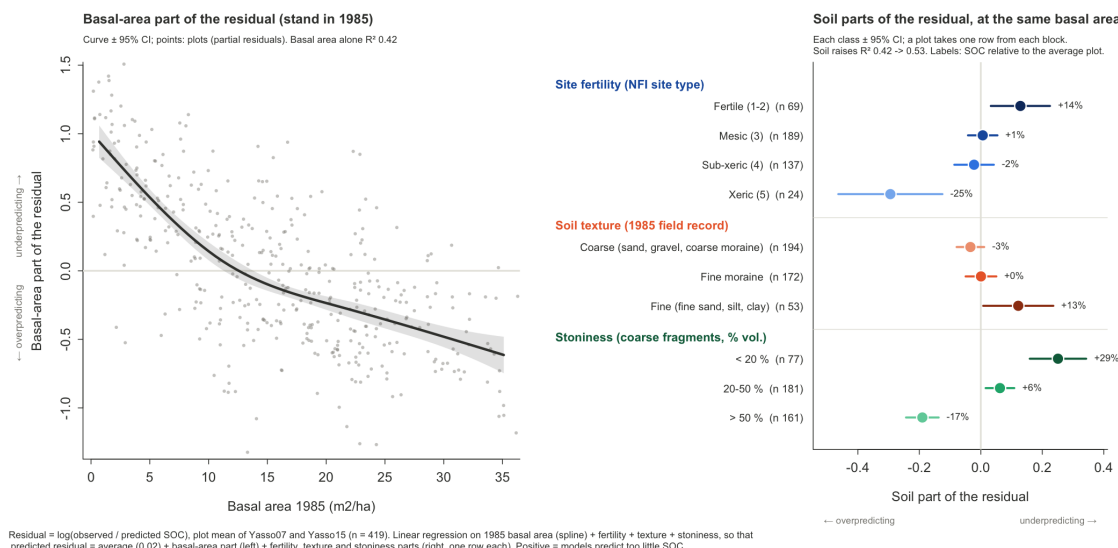


Figure 19: F11_stand_and_soil

3.0.14.3 Side 1 — the 1985 stand. Sparse 1985 stands hold $\sim 2.5\times$ the predicted carbon, dense ones $\sim 0.5\times$; basal area alone $R^2 = 0.42$. The site parts are minor and ecologically sensible: fertile +14%, xeric -25%, fine-textured +13%, stone-poor +29%, very stony -17% (R^2 0.42 $\rightarrow$ 0.53). Nothing else adds (defoliation, damage, lichens, management, climate, wetness, N deposition, 2023 understorey cover — the latter converted to litter adds only 13% to the tree input and is unrelated to the residual).

3.0.14.4 Side 2 — time since the last regeneration cut (the development-class lead made explicit). -20% in the first 5 years, +55% at 20–30 years, about zero or below in old stands; significant beyond basal area ($p = 4 \times 10^{-11}$; dated cuts only $p = 9 \times 10^{-4}$) but small in variance. Split by 1985 basal area, the hump lives **only** in stands $< 10 \text{ m}^2 \text{ ha}^{-1}$ in 1985 (+50% $\rightarrow$ +150%). ► [F11b_time_since_cut](#)

3.0.14.5 The connection — one finding, a wrong starting state. Sparse 1985 stands were young (median 19 yr; two thirds aged 10–30 = the hump window). In the same plots the excess **shrinks**: +150% (first sampling) $\rightarrow$ +70% (2006) $\rightarrow$ +30% (2024); dense stands -36% $\rightarrow$ -25%. Largest at the start and fading at about the models' turnover pace = an initial-state error, not an input missing throughout. Cause: the 1917–1985 input path has one national shape scaled to each plot's 1985 input, so the start knows nothing of the plot's history. **The thesis one level down.**

3.0.14.6 ~ The pointer — pioneer vegetation. The excess carbon of the 1985-young stands was built before 1985, during early stand development: inherited from the previous stand, and/or

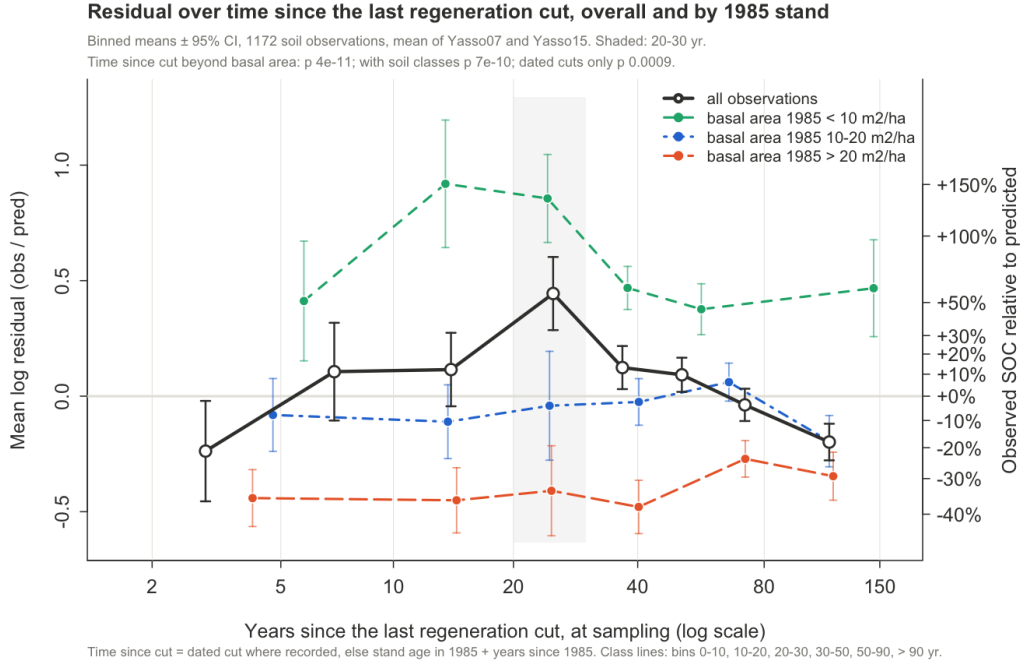


Figure 20: F11b_time_since_cut

supplied by early-succession vegetation (pioneer birch, rowan, aspen, willow below the tally threshold and removed in cleaning; a productive field layer) that no tree-based litter product records. Evidence that the early-succession input is large: Kulmala et al. 2011, Peichl et al. 2022, Grelle et al. 2023, Kolari et al. 2004; that it escapes the inventory and standard biomass models: Repola et al. 2024, Rana et al. 2023 (Mehtätalo), Dahlgren Lidman et al. 2024. **Caveat to keep:** post-1985 dated cuts do not (yet) show a comparable excess, so inheritance and pioneer input are not separated by our data. Next step, briefly: a plot-specific start, the cut dated by the 1985 stand age.

4 Conclusions

Annotation (p22): “The policy deliverable should not be a real conclusion, it is more a nice bonus product. The conclusions ... are more about the opening stories, their resolutions.”
Rebuilt exactly that way: the conclusions below answer the three opening questions in the order of the weight budget, and the deliverable is demoted to a closing paragraph.

4.0.0.1 A transient, calibrated start reproduces the observed dynamic with a starting deficit that the conventional start cannot carry. The observed accumulation and its levelling-off are recovered by all six models at the level, and bracketed by them in rate. [corrected 2026-09-16] A balanced start can also rise and level off, but only by attributing the whole of it to litter changes inside the window; the two starts differ in the product $J \times \tau$ they imply, and that difference is the headroom of the fourth conclusion.

4.0.0.2 Reproducing it requires more litter input, and faster turnover, than the published parameterisations carry. These are one displacement along a trade-off the data cannot resolve, and the most plausible reading is not that the models are wrong but that the input flux is

incomplete — with belowground and non-tree sources the candidates.

4.0.0.3 Where the calibration lands on that trade-off reaches the temperature sensitivity, and through it the projection, but only for some structures. A single climate modifier ties turnover to temperature sensitivity, so the starting-state assumption that sets the position on the trade-off sets the warming response with it; pool-specific modifiers do not. Models of the second kind are more robust to an assumption that cannot currently be tested.

4.0.0.4 There is headroom, and taking it would cost something. Five of the six models put today’s soil 28–49% below the stock its present litter input would eventually support, which is where Liski et al. (2006) also put it. That is space for further accumulation — but realising it means holding or raising the carbon actually returned to the soil, over decades, with the management and economic consequences that implies. **And it postpones saturation rather than removing it:** the same models that say there is room also say the room is finite.

Annotation (p7): “*there would be space for some more accumulation? But we need to manage forests in a way that more C is returned to the soil, with all its economic consequences ... (and it would be just postponing it, it will never go on indefinitely).*” **Adopted as a conclusion, with the number attached.** This is also the honest limit of the policy reading: a finite ceiling is not a permanent sink.

4.0.0.5 And a large, recoverable part of the plot-scale variance is invisible to every model here. It is a well-posed question for the next study rather than a limitation of this one.

4.0.0.6 Bonus product, not conclusion: the maps. The calibrated ensemble yields gridded SOC and its uncertainty for Finland, which is useful to the inventory and is offered as a deliverable rather than as the paper’s finding. ► [ENSEMBLE_SOC_mean](#)

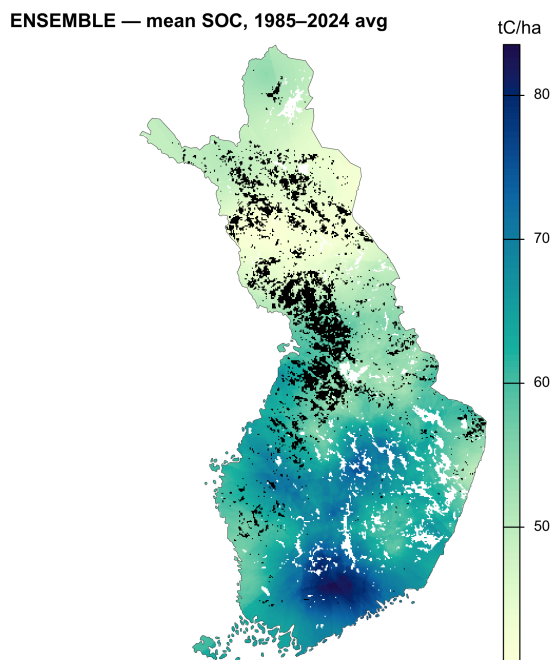


Figure 21: ENSEMBLE_SOC_mean (NextGenC deliverable)

Open decisions and what is still missing

4.0.0.7 Structural. Whether Results and Discussion stay combined (currently yes). Whether F3b is built or the NextGenC panel suffices. Whether the arc diagram’s thread vocabulary is retired.

4.0.0.8 The σ_{input} prior anchor — settled, and it does not threaten the finding. Arm B’s 1.08 mixes two litter products; the defensible centre is ~ 1.21 , and Liski gives 1.27. **Decision: stay on arm B**, state the arithmetic in the Methods, revisit only if a referee asks.

Annotation (p24): “*Here I am not sure at all ... we saw that with Arm A results do not change much, our sigmas are still ways higher than the data would justify, right? So there would still be something missing.*” **Correct, and I was wrong — measured on the two runs we already have.** Moving the prior centre $1.08 \rightarrow 1.27$ (arm A, 20260820_1554*) moves the posterior only about a third as far: Yasso15 $1.54 \rightarrow 1.62$, TP2 $1.73 \rightarrow 1.87$, SP1 $1.27 \rightarrow 1.35$. The posterior sits 1.28 to $1.50\times$ **above the prior centre in arm A**, as it sits 1.17 – $1.60\times$ above it in arm B. **The likelihood, not the prior, is setting σ_{input}** ; the displacement survives either anchor, and my claim that a higher centre would shrink the finding was false.

⊢ REFS ours, both production runs; posterior sensitivity to the prior centre ≈ 0.3 – 0.5

4.0.0.9 Literature reviews — now drafted, with what remains. All eight topics are written up in literature/NN_topic/README.md (evidence table, draft text, gaps) with a cross-cutting MISSING_FLUX_BUDGET.md. Still blocking: ☐ Peltoniemi et al. 2006 (before the spin-up quote is used as a foil); ☐ the turnover/exudation split inside Bolinder’s 65%; ☐ Hansson et al. on fine-root flux weighting; ☐ Clemmensen et al. 2013 read in full; ☐ primary sources for CBM-CFS3 and Century.

4.0.0.10 Three checks before their numbers can be quoted. ☐ The area and depth basis of Liski et al.’s $6.1 \rightarrow 6.3 \text{ kg C m}^{-2}$, and of their 6.1 kg C m^{-2} in 1922 as a candidate anchor for the still-unsourced 1917 stock floor. ☐ The component key of our own litter product (the belowground share, on which the rhizodeposition estimate rests) — Boris Tupek. ☐ Kumpu et al.’s depth basis against our whole-profile target.

4.0.0.11 Still not started. Citations and a .bib; title and abstract; the coauthor block; figure placement in the manuscript.

4.0.0.12 Unresolved from earlier sessions. The 1917 stock floor is still an unsourced placeholder, and the relaunch of the six calibrations after the Fortran guard change has not been done.